

3^n SEGMENTS AND 3^n NATURAL CLASSES, EACH OF CARDINALITY 3^k

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I defend the following three claims under the assumptions of Logical Phonology: Universal Grammar makes available an inventory of 3^n segments; there are 3^n natural classes; and the cardinality of each natural class is 3^k , $1 \leq k \leq n$.

Keywords: Segments, Natural classes, Features, Logical Phonology, Underspecification

INTRODUCTION

UNDER the fairly standard assumptions of Logical Phonology (LP, e.g., Bale *et al.* 2014; Bale & Reiss 2018; Reiss 2021; Volenec & Reiss 2020; Leduc *et al.* 2024; Gorman & Reiss 2025) outlined below, I defend the following three claims:

- (1) Three claims about segments in Logical Phonology
 - i. The primitives supplied by Universal Grammar make available an inventory of 3^n phonological segments.
 - ii. Each natural class is defined by the features that constitute one segment, so there are 3^n natural classes.

- iii. The cardinality of each natural class is 3^k , $1 \leq k \leq n$, where k is the number of features for which the class is *not* specified.

I now lay out the LP assumptions for formalizing segments and natural classes, illustrating with a schematic universal phonology, UG_{toy} . First assume that UG_{toy} provides a set $\mathcal{F}$ containing just two features G and H, along with a set of values $V = \{+, -\}$. Further structure to the feature inventory, such as Feature Geometry (e.g., Sagey 1986) is not assumed—justification for rejecting Feature Geometry is provided by Reiss (2003).

(2) Primitives of UG_{toy}

- Universal feature set: $\mathcal{F} = \{G, H\}$
- Universal set of values: $V = \{+, -\}$

The traditional generative view (e.g., Chomsky & Halle 1965) that features are innately provided by Universal Grammar is defended at length by Reiss & Volenec (2022).

Valued features are ordered pairs of the form ωF , where $\omega \in V^1$ and $F \in \mathcal{F}$. UG_{toy} in (2) yields, by the free combination of values and features (the cross product $V \times \mathcal{F}$), a set $\mathbf{F}$ of four valued features $\{-G, -H, +G, +H\}$.

(3) Universal set of valued features: $\mathbf{F} = V \times \mathcal{F} = \{-G, -H, +G, +H\}$

Abstracting away from other structures, such as timing slots, we then define a segment as a *consistent* subset of $\mathbf{F}$. A set ς is consistent if, for any F in $\mathcal{F}$, ς does not contain both $+F$ and $-F$ as members.

Underspecification is possible in this model because there is no stipulation that seg-

¹I use ω as a metalanguage variable, rather than α , which I reserve for use as a variable within grammar, *pace* McCawley (1999) who treats Greek letter variables as metalanguage variables, and not available to the grammar. In other words, a rule in a mental grammar can refer to, say, agreement of segments with respect to αF , so there must be at least two tokens of α in any rule; in contrast, a phonologist can refer to a segment that is specified $\omega_1 G$, $\omega_2 H$, etc., with just a single token of each ω_i .

ments be *complete*; there is no requirement that a segment have a valued feature for each feature in $\mathcal{F}$. The model thus stipulates the existence of exactly two values of V , but derives the three possibilities of +, – and ‘absent’ by the lack of a stipulation of completeness. This ternary use of a binary system is by no means a sloppy idiosyncrasy of LP and related phonological models—for some discussion of such matters with respect to constructivist logic and computer science see the blog entry ‘How many is two?’ at <https://math.andrej.com/2005/05/16/how-many-is-two/>.

1 HOW MANY SEGMENTS ARE THERE?

We can then define the universal segment inventory Σ as the set of consistent subsets of $\mathbf{F}$, that is, the set of subsets of $\mathbf{F}$ that do not contain opposite valued features, like +G and –G. Each of these segment sets, then, contains –G, +G or neither, and contains –H, +H or neither. In other words, there are three possibilities for each feature in a segment: “+”, “–” and “absent”. With just two features, then, Σ has $3^2 = 9$ members. The nine segments definable with UG_{toy} are given in (4) as members of the set Σ , numbered from 0 to 8.² If another UG instead provided twenty features then Σ would have 3^{20} , or about 3.5 billion, segments, which shows that a fairly conservative UG can still offer a rich variety of segments via the combinatoric explosion that Gallistel & King (2009) identify as a desirable trait in a representational system (Reiss 2012; Matamoros & Reiss 2016).

(4) Universal segments in a two feature system

$$\Sigma = \left\{ \begin{array}{cccccccccc} \{ \} & \{-H\} & \{+H\} & \{-G\} & \{-G, -H\} & \{-G, +H\} & \{+G\} & \{+G, -H\} & \{+G, +H\} \\ \varsigma 0 & \varsigma 1 & \varsigma 2 & \varsigma 3 & \varsigma 4 & \varsigma 5 & \varsigma 6 & \varsigma 7 & \varsigma 8 \end{array} \right\}$$

Each feature added to $\mathcal{F}$ will triple the number of segments in Σ : for example, if we add

²The order of the nine segments from 0 to 8 corresponds to the ternary numbers from 0 to 22 using H for the *ones* column and G for the *threes* column. This scheme just provides a standard way of enumerating all the segments in a manner that is extensible for any number of features. This is an example of the Wolfram Code (https://en.wikipedia.org/wiki/Wolfram_code).

a feature J, then, say, ς_8 above will map in the new Σ to $\{+G, +H, -J\}$, $\{+G, +H, +J\}$ and $\{+G, +H\}$. Our first claim (1)i is now clear: there are always 3^n segments in Σ where n is the number of features provided by UG.

2 HOW MANY NATURAL CLASSES ARE THERE?

Another major assumption of LP is the widely accepted idea that natural classes “can be specified by a single conjunction of features” (Halle & Clements 1983:9) or “conjunctions of properties” (Odden 2013:159). We assume that such a coherent notion of natural classes is a necessary component of a theory of rules (or any other computational system), despite the fact that the literature is rife with statements that indicate a lack of appreciation for the intimate relation between natural class logic and rule syntax. For example, Hayes (2009) says that in rules, natural classes occur “in most instances” (p. 43) or as a “pervasive tendency” (p. 71). Kornai (2008), assuming a language with a segment inventory that includes at least $/p, t, b, d/$, says that “Phonologists would be truly astonished to find a language where some rule or regularity affects p , t , and d but no other segment”—presumably he has b in mind. Mielke (2008:9) claims that “unnatural classes are widespread” in rules. The idea is that some rules do *not* refer to natural classes—but such a claim is empty since it is not accompanied by a theory of what is a possible rule. Our position is that a definition of natural classes is a necessary component of a theory of rules—natural classes define rule targets and environments.

A third assumption for this discussion is that we are concerned with intensionally defined universal natural classes, that is, subsets of the universal segment inventory Σ determined by the primitive system of features and values sketched above. This approach contrasts with most other discussions which conflate the issue of universally defined classes and the “image” of such classes in a given language, such as Wheeler (1972) who examines subsets of the twenty-nine segments of a Catalan dialect to determine

which are “more” natural than others. To further clarify our approach, consider that in English the set of segments {p, b, m} would typically be called a natural class, but the set {p, m} would not be; however, in Cree the set {p, m} may sometimes be called a natural class since the language has no /b/, and so the set with just two members constitutes the class of labial consonants *in that language*. In contrast, we define classes universally and intensionally, e.g., “the labial consonants,” without regard to the extension of such a class in a given language’s segment inventory at some (or all) levels of representation.

Under LP assumptions, segments are *sets of valued features*, and thus sets of segments are *sets of sets of valued features*. We need a definition of natural class that distinguishes natural sets, or classes, of segments from other sets of segments. To achieve this, we adapt from Bale & Reiss (2018) a formalization of the traditional notion of natural classes defined by a conjunction of features:

- (5) **Definition of natural class:** A subset Q of the universal segment inventory Σ is a natural class if and only if there exists a consistent set of valued features $\mathcal{C}$ such that $Q = \{q \mid q \supseteq \mathcal{C}\}$.

In other words, each segment q in a natural class is a superset of some set of features $\mathcal{C}$. Unlike Bale and Reiss, and most discussions in the literature, I do not require that each q be an actual element of the segment inventory of the particular language under analysis—as noted, we are interested in universal natural classes.³

The second of our claims (1)ii now becomes trivially obvious. Each natural class is defined in terms of a consistent set of valued features $\mathcal{C}$. Since we have already shown how to enumerate all such sets of valued features—there are nine of them for UG_{toy} in (4) and 3^n in general—there are always 3^n natural classes of segments. It is worth pointing out that for UG_{toy} , these nine classes are a small proportion of the arbitrary

³We follow (Bale & Reiss 2018:Unit 45) in supposing that natural class formulae in rules arise in the course of acquisition via a process of generalized intersection—this allows for the possibility of singleton natural classes.

sets of segments one gets by taking subsets of Σ : for a given subset, each segment is either ‘in’ or ‘out’, so there are $2^9 = 512$ possible subsets of Σ . Only nine of these member sets of the *powerset* (the set of all subsets) of Σ are natural classes. For example, for UG_{toy} the class $\{\varsigma 3, \varsigma 4, \varsigma 5\}$ is natural, but the class $\{\varsigma 2, \varsigma 3, \varsigma 5\}$ is not.

3 THE CARDINALITY OF A NATURAL CLASS

Let’s now explore what values can correspond to the cardinality of a natural class Q .

3.1 CLASS OF CARDINALITY 0?

Since $\mathcal{C}$ in the definition in (5) is a consistent set of valued features, $\mathcal{C}$ itself is a member of Σ , the set of definable segments. Moreover, since $\mathcal{C}$ is a superset of itself (like all sets, according to a basic theorem of set theory), there will be some q in Q equal to $\mathcal{C}$. It thus follows from the definition that Q cannot be the empty set, or equivalently, that the empty set is not a natural class of segments— Q must contain at least the segment $\mathcal{C}$.⁴ So, a natural class cannot have a cardinality of 0. Since there is no power of 3 equal to zero, this is consistent with our claim in (1)iii about the cardinality of natural classes.

3.2 CLASS OF CARDINALITY 3^n ?

Consider now the case where $\mathcal{C} = \{ \}$, corresponding to a completely unspecified segment. What is the cardinality of the class defined using this value for $\mathcal{C}$? In this case, the natural class defined is $\{q \mid q \supseteq \{ \} \}$, that is, for UG_{toy} , the class defined by the features of $\varsigma 0$.

⁴Thanks to Rim Dabbous for making me see this. It may be useful to clarify some issues here. First, as just shown, the empty set *of segments* is a subset or subclass of Σ , but it is not a natural class, given the definition in (5). Second, the empty set *of valued features*, which we turn to just below, is a *member* of Σ . However, set theory does not allow us to distinguish the two empty sets we just referred to, because another theorem of set theory tells us that the empty set is unique—there is only one empty set. Thus, the empty set of planets and the empty set of toes are the same set. This means that the set theory type difference between *sets of valued features* and *sets of sets of valued features* (i.e. sets of segments) is erased when dealing with the empty set.

Of course, this is the set of *all* segments, the universal segment inventory Σ , since every set is a superset of the empty set, by yet another fundamental theorem of set theory. As noted above, there are 3^n possible segments, and thus, that is the cardinality of the universal segment set, Σ . So, with n features, there is a natural class which has 3^n members.

3.3 CLASSES OF CARDINALITY 1?

A recurring confusion in the literature is that natural classes cannot be singleton sets. For example, Davenport & Hannahs (2010:91-92) say that “Natural classes may consist of any number of sounds, from two to many...”. We have seen that our definition of natural classes ensures that every natural class must have at least one member, namely the segment whose feature set is the same as the feature set used to define the class. As we saw above, the segment $\mathcal{C}$ will always be a member of the class $\{q \mid q \supseteq \mathcal{C}\}$, so every natural class has at least one member. Now if $\mathcal{C}$ is fully specified, such that for each feature it contains either $+F$ or $-F$, then no other segment $\mathcal{D}$ can be a superset of $\mathcal{C}$, since $\mathcal{D}$ will either have the opposite value for F , or lack a value for F altogether. So, there are classes of cardinality $3^0 = 1$, with each such class containing a fully specified segment. For UG_{toy} these are the classes defined by the features of a fully specified segment, $\varsigma_4, \varsigma_5, \varsigma_7$ or ς_8 . Since we know that there are rules that target only one specific segment—onset glottal stop deletion in Suva Rotuman (Fimone 2020), to cite a random example—this is clearly a desirable outcome, and the exclusion of singleton natural classes by Davenport and Hannahs must be rejected.

3.4 CLASSES OF CARDINALITY 2?

We can also see that no *two* segments $\mathcal{C}, \mathcal{D}$ can constitute a natural class. For ease of exposition consider the two situations in which two segments are maximally similar. First consider the case where there is at least one feature F such that $\mathcal{C}$ is specified ωF

and $\mathcal{D}$ has no specification for F , but is otherwise identical to $\mathcal{C}$. If a natural class contains both $\mathcal{C}$, $\mathcal{D}$, then the set of features that define the class must not contain any specification for F . Whatever that set is, $\mathcal{C}$ and $\mathcal{D}$ must be a superset of that set. Therefore, the set, $\mathcal{E}$, that is identical to $\mathcal{C}$ aside from containing $-F$, will also be a superset of that set. In other words, the natural class must have at least the three members $\mathcal{C}$, $\mathcal{D}$ and $\mathcal{E}$. For concreteness, this means that, say, the set $\{t, D\}$, where $/D/$ is a coronal stop unspecified for VOICED, is not a natural class, since $/d/$ must be in any class that contains the other two.

Consider now the sets $\mathcal{C}, \mathcal{E}$ whose members contain opposite values for some F . Again, the natural class specification must exclude F , and thus $\mathcal{C}, \mathcal{E}$ as well as $\mathcal{D}$ must be in the class. In other words, no natural class can contain both $/t/$ and $/d/$ but not $/D/$.

In the simplest cases of the above examples where only one feature F is involved, we see that both $\mathcal{C} \cap \mathcal{D}$ and $\mathcal{C} \cap \mathcal{E}$ yield $\mathcal{D}$, which is of course a subset of all three segments. At this point we should remember that the set of Cree segments $\{p, m\}$, mentioned above, does not constitute a (universal) natural class, but is merely the image, or extension, of that class in a particular language. So, a natural class cannot contain two members, which is expected, since there is no power of 3 equal to 2. As we have seen, natural classes with three members *are* possible: the set $\{t, D, d\}$ is one example.

3.5 THE GENERALIZATION

Each natural class is specified by some consistent set of valued features. Assume a class is specified for the valued feature ωF , then every member of the class contains ωF —there is only one possibility. In the case of a fully specified segment, the natural class contains just one member because there is one choice for each F .

On the other hand, if a class is not specified for F , then all three choices, $+F$, $-F$ and “absence of F ”, are possible for members of the class. The cardinality of a natural class will be $1^j \times 3^k$ where j is the number of features for which the class is specified

and k is the number of features for which the class description is not specified, thus $j + k = n$. Since $1^j = 1$ no matter what j is, and since 3^k is a power of 3, the product $1^j \times 3^k = 3^k$ is always a power of 3. The value of k lies between (a) n , the case where the natural class description is fully unspecified, yielding Σ , the universal set of segments, whose cardinality is 3^n ; and (b) 0, the case where the natural class description contains all the valued features of a fully specified segment, yielding a singleton natural class of cardinality $3^0 = 1$. The validity of our third claim (1)iii is now apparent: every phonological natural class in a system with n features is an integer power of 3, 3^k , where $0 \leq k \leq n$.

DISCUSSION

This understanding of the content of natural classes has practical implications for phonological analysis, including the use of underspecification and the precise formulation of rules. For example, in a language with /t, D, d/ no rule should be posited that is intended to target any two of those segments but not the third, if all three appear in the language at relevant derivational levels.

Also, a rule that targets singleton natural classes of fully specified segments like {t} is possible, but rules that target only classes containing underspecified segments like /D/ are not possible (intensionally). Such issues have been long noticed, starting at least with Lees (1961:14), who spoke of “the constraint on phonological rules that they never be required to identify unspecified, or zero, features” and continuing over intervening decades in discussion of the contrast between feature-filling and feature-changing processes. The apparent targeting of segments like /D/ to the exclusion of /t/ and /d/ requires a means of formulating strictly feature-filling processes. The Logical Phonology (LP) literature (e.g., Bale *et al.* 2014; Bale & Reiss 2018; Reiss 2012) is a recent approach to this issue relying on unification-based rules. Of course, other phonological frameworks

make other assumptions—for example, the use of privative features requires a different approach to feature-filling rules.⁵

It is perhaps of interest to point out that the results here involving powers of three can be easily adapted to a model of UG that requires completeness, i.e., that does not allow for underspecification. In such a case, the number of segments will be 2^n , since each segment will be specified either +F or −F for each feature. The number of natural classes, however, remains 3^n , because every consistent set of valued features provides a partial description to define a natural class—in other words, every subset of a fully specified segment can define a natural class, and there are 3^n possibilities for these, as above. Using the features of UG_{toy} , but requiring full specification, the only segments in Σ are $\varsigma 4, \varsigma 5, \varsigma 7, \varsigma 8$, but $Q = \{q \mid q \supseteq \{+G\}\}$ is still a possible natural class—in this case with just two members, $\varsigma 7$ and $\varsigma 8$. Natural classes in a system without underspecification must have cardinality 2^k , where k is the number of features which do not appear in the natural class characterization, for reasons parallel to the argument above—for each feature left out of a natural class description, the number of segments doubles. If $k = n$, once again the relevant natural class is the same as Σ , the universal set of segments, in this case of cardinality 2^n .

It is worthwhile to note the benefit of adopting the Logical Phonology view that insights arise by treating segments as sets and leveraging the results of set theory. We were able to appeal to three theorems of basic set theory to better understand natural classes:

- Every set is a superset of itself.
- Every set is a superset of the empty set.
- There is only one empty set.

⁵Useful discussions of variation features systems include Gorman (2026); Nelson (2022); Kim (2002) and references therein.

This use of set theory exemplifies Chomsky’s 2005 “third factor” mode of explanation by calling upon “[p]rinciples not specific to the faculty of language” in order to understand which I-languages can be attained, since the results follow from mathematical logic, once we treat segments as sets of valued features, and not from strictly phonological considerations (real or hypothesized) such as contrastiveness, markedness or constraints on abstractness.

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